

Mehrdad Kashefi

Contact Info	Western Interdisciplinary Research Building, London, ON, Canada N6A 3K7 mkashefi@uwo.ca mkashefi.com	
Education	Western University , London, ON, Canada <i>Ph.D. Neuroscience</i> Thesis Title: “Neural Basis of Sequential Actions” Advisors: Dr. Andrew Pruszynski, Dr. Jörn Diedrichsen Defense date: November 7th, 2025	2020 - 2025
	Iran University of Science and Technology (IUST) , Tehran, Iran <i>M.S. Electrical Engineering - Bioengineering</i>	2017 - 2020
	Lorestan University , Lorestan, Iran <i>B.S. Electrical Engineering - Electronics</i>	2013 - 2017
Awards	COSYNE Presenters Travel Grant–USD \$500	2025
	Neural Control of Movement (NCM) Travel Award	2024
	International Ontario Graduate Scholarship–CAD \$15,000	2023
	Best Oral Presentation at Neuroscience Research Day, Western University	2023
	Neuroscience Program Travel Award, Western University–CAD \$500	2022
Publications		
Preprints	[1] Kashefi M. , Michaels J., Kersten R., Lau J., Diedrichsen J., Pruszynski J.A. “Compositional neural dynamics during reaching”, bioRxiv 10.1101/2025.09.04.674069	
Published	[9] Michaels J., Kashefi M. , Zheng J., Codol O., Weiler J., Kersten R., Lau, J., Gribble P., Diedrichsen J., Pruszynski J.A. “Sensory expectations shape neural population dynamics in motor circuits”, Nature, 2025	
	[8] Kashefi M. , Diedrichsen J. [†] , Pruszynski J.A. [†] “Motor sequence learning involves better prediction of the next action and optimization of movement trajectories”, Journal of Neuroscience, 2025 († := Co-senior authors)	
	[7] Shahbazi M., Ariani G., Kashefi M. , Pruszynski J.A., Diedrichsen J., “Neural correlates of online action preparation”, Journal of Neuroscience, 2024	
	[6] Kashefi M. , Reschechtko S., Ariani G., Shahbazi M., Alice Tan, Diedrichsen J., Pruszynski J.A. “Future movement plans interact in sequential arm movements”, eLife.94485.1,2024	
	[5] Chung B., Zia M., Thomas K., Michaels J., Jacob A., Pack A., Williams M., Nagapudi K., Teng L., Arrambide E., Ouellette L., Oey N., Gibbs R., Anschutz P., Lu J., Wu Y., Kashefi M. , Oya T, Kersten R, Mosberger A., O’Connell S, Wang R., Marques H, Rita P. Mendes A., Lenschow C., Kondakath G., Kim J., Olson W.,Quinn K., Perkins P., Gatto G., Thanawalla A., Coltman S., Kim T., Smith T., Binder-Markey B., Zaback M., Thompson C., Giszter S., Person A., Goulding M., Azim E., Thakor N., O’Connor D., Trimmer B., Lima S., Carey M., Pandarinath C., Costa R., Pruszynski A., Bakir M., Sober S. “Myomatrix arrays for high-definition muscle recording”, eLife.88551.1, 2023	
	[4] Codol O., Michaels J. A, Kashefi M. , Pruszynski J.A., Gribble P. “MotorNet: a Python toolbox for controlling differentiable biomechanical effectors with artificial neural networks”, eLife.88591.1, 2023	

- [3] Codol O., **Kashefi M.**, Forgaard C.J., Galea J.M., Pruszynski J.A., Gribble P. “Sensorimotor feedback loops are selectively sensitive to reward”, eLife 12:e81325, 2023
- [2] Ahmadi A.*, **Kashefi M.***, Shahrokhi H, Nazari M.A., “Computer aided diagnosis system using deep convolutional neural networks for ADHD subtypes”, Biomedical Signal Processing and Control, 2021 (* := equal contribution)
- [1] **Kashefi M.**, Daliri M.R., “A stack LSTM structure for decoding continuous force from local field potential signal of primary motor cortex (M1)”, BMC bioinformatics, 2021

Conference

Talks

Kashefi M., Michaels J. A, Diedrichsen J., Pruszynski J.A., “Motor cortical dynamics during reaching connect posture-specific attractors”, Neural Control of Movement (NCM), Panama City, Panama, 2025

Kashefi M., Diedrichsen J., Pruszynski J.A., “A compositional solution for sequence learning”, Neural Control of Movement (NCM), Dubrovnik, Croatia, 2024 (Online)

Kashefi M., Diedrichsen J., Pruszynski J.A., “Is motor learning merely anticipation?”, Neuroscience Research Day (NRD), London, Canada, 2023

Posters

Kashefi M., Michaels J. A, Diedrichsen J., Pruszynski J.A., “Motor cortical dynamics during reaching connect posture-specific attractors”, Cosyne, Montreal, Canada, 2025

Kashefi M., Diedrichsen J., Pruszynski J.A., “Is there more to sequence learning than better anticipation?”, Neural Control of Movement (NCM), Victoria, Canada, 2023

Kashefi M., Ariani G., Diedrichsen J., Pruszynski J.A., “Planning multiple future actions in sequential reaching”, Neural Control of Movement (NCM), Dublin, Ireland, 2022

Invited talks

Kashefi M., “Compositional neural dynamics during reaching”, Shine Lab, University of Sydney, October 2025 (Online)

Kashefi M., “Motor cortical dynamics during reaching connect posture-specific attractors”, Chinese institute for brain research, July 2025 (Online)

Kashefi M., “Future movement plans interact in sequential arm movements”, Motor control Group, West Virginia University, December 2024 (Online)

Teaching

Mentorship

Armin Panjehpour (PhD, Western University, 2023-)

Alice Tan (BSc, Western University, 2022-2023)

Workshop

Introduction to Machine Learning, Brainhack Western 2021

TA

Physiology and Pharmacology Laboratory, Western University 2021
Computational Neuroscience, IUST 2019-2020
Electronic Circuits, IUST 2018

Professional Service

Peer review

Nature Human Behaviour (#1) — eLife (#2)

Outreach	Mentorship Committee, Society of Neuroscience Graduate Students, Western Mentoring first-year graduate students in the neuroscience program.	2022
	Thames Valley Science and Engineering Fair, London, ON, CA Judge for Grade 4-12 Science Engineering Fair	2021-2023
	Python Camp, IUST Introductory python course for middle school students	2019
	Executive Committee Member, ICBME , IUST 26th International Iranian Conference on Biomedical Engineering (ICBME)	2019
Referees	Dr. Andrew Pruszynski Associate Professor, Canada Research Chair in Sensorimotor Neuroscience Western University Email: andrew.pruszynski@uwo.ca	
	Dr. Jörn Diedrichsen Professor, Western Research Chair for Motor Control and Computational Neuroscience Western University Email: jdiedric@uwo.ca	
	Dr. Jonathan Michaels Assistant Professor York University Email: jmichae@yorku.ca	